



TOSHKENT SHAHRIDAGI INHA UNIVERSITETI

INHA UNIVERSITY IN TASHKENT

–1 POINTS FOR EACH FIELD LEFT EMPTY OR FILLED WRONG

ID NUMBER:

FIRST NAME: LAST NAME:

STUDENT SIGNATURE: ROOM NUMBER

FINAL EXAMINATION

COURSE NAME: COURSE NUMBER:

EXAMINATION DURATION:

ADDITIONAL MATERIALS ALLOWED TO BE USED:

SPECIAL INSTRUCTIONS

(1) FOLLOW ANY INSTRUCTIONS GIVEN BY PROCTORS.
(2) ONCE YOU ARE ALLOWED TO OPEN THE EXAMINATION BOOKLET, YOU MUST PUT YOUR **ID** ON EVERY ODD PAGE
(3) READ THE INSTRUCTION CAREFULLY!
Sometimes it is asked to provide JUST an answer, sometimes **FULL** solution.

- Final answers must be written by **only blue, non-erasable pen**.
- Do NOT use highlighters or correction pen.
- IUT's Examination Rules and Regulations apply.

Do NOT open the examination paper until directed to do so!

Do NOT fill	P.1	P.2	P.3	P.4	P.5	Total
max points	8	8	8	8	8	40
your score						

1. (8 points) Let $a_n = 49a_{n-2}$ be a recurrence relation.

a) (1 point) what is the degree of this relation?

b) (4 points) find all solutions of the recurrence relation.

c) (3 points) find the solution of the recurrence relation with initial conditions $a_0 = 0, a_1 = 14$.

2. (8 points) Let $A = \{a, b, c, d\}$ and $R = \{(a; a), (a; d), (b; b), (b; c), (c; c), (c; d), (d; a), (d; b), (d; d)\}$

a) (2 points) represent this relation with a matrix.

b) (1 point) is this relation reflexive? (yes/no)

c) (1 point) is this relation symmetric? (yes/no)

d) (2 points) is this relation antisymmetric? (yes/no)

e) (2 points) represent this relation with directed graph.

3. (8 points) Let $(\{2, 3, 9, 10, 12, 30, 50, 120\}, |)$ be a poset.

a) (4 points) construct the Hasse diagram for this poset.



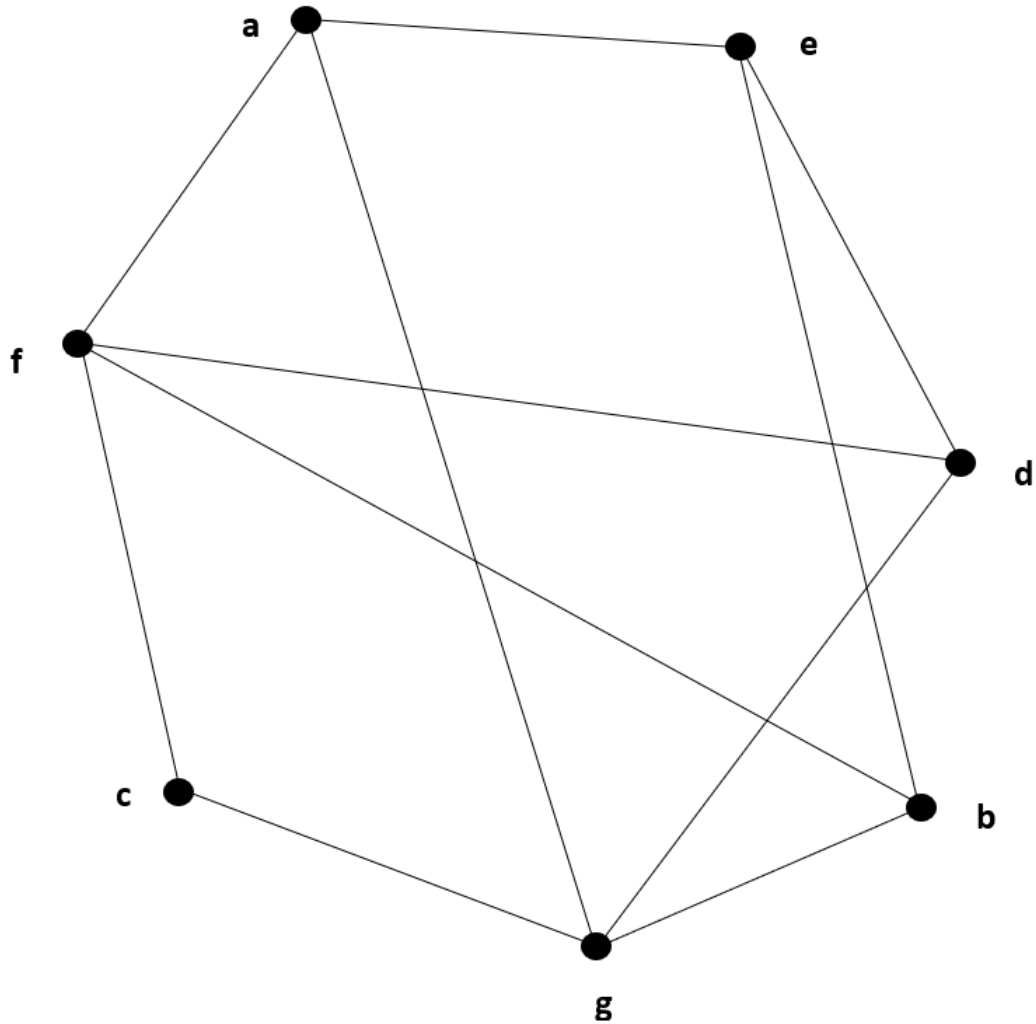
b) (2 points) list the maximal elements of this poset.



c) (2 points) list the minimal elements of this poset.

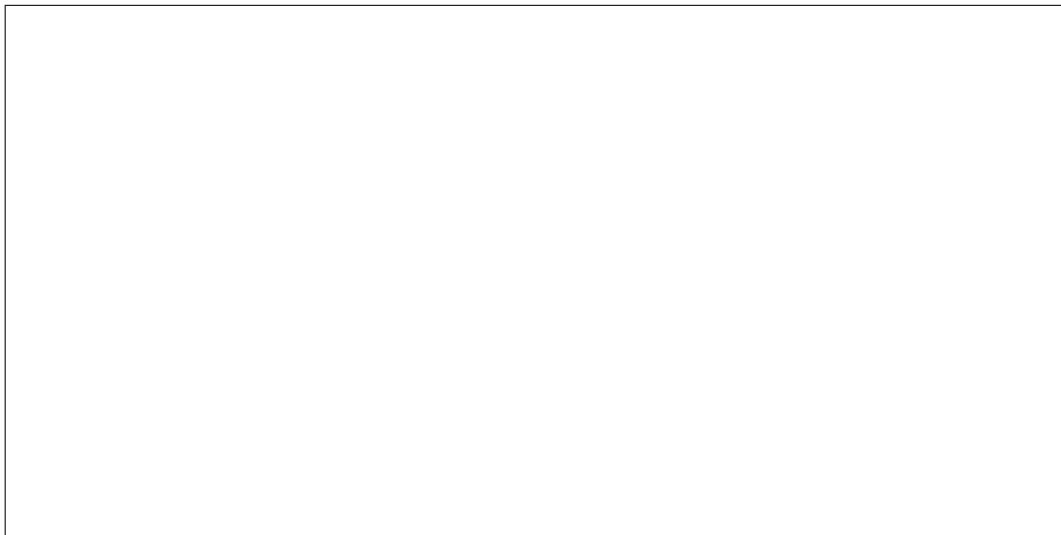
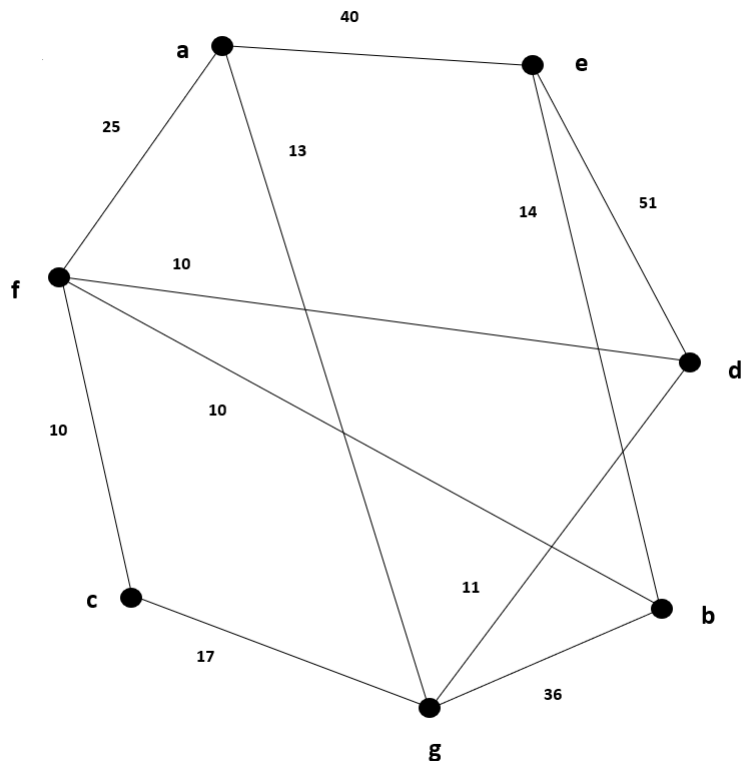


4. (8 points) Consider the following graph.



a) (3 points) find a bipartition of this graph.

b) (4 points) draw a minimum spanning tree of this graph, if there are weights.



c) (1 point) what is the weight of the minimum spanning tree?

5. (8 points) Using a principle of mathematical induction show that $7^n - 3^n$ is divisible by 4 for any $n \geq 0$.

